Neeraj Sahu

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Skills

Technical Skills: Performance Validation, Wind Farm Control, Certification

Languages: Python, MATLAB, Bash, R

Simulation & Modelling: BHaWC, FLORIS, PyMC (Bayesian Modeling), State-Space Models

Standards: IEC 61400

Tools: Git, LaTeX, Jira

Experience

Siemens Energy Wind Power Solution Designer - R&D *October 2024 – Present*

- **Quarterly Controller Certification:** Owned the planning and execution of quarterly certification documentation for the D8, SG5X, and D3 fleet, supervising with a senior consultant and intern to meet deadlines.
- Aligned with multiple stakeholders such as Solution Owners, Product Owners, and Tool Developers.
- Evangelized a comprehensive understanding of the certification process, including the IEC 61400 standard and its application to wind turbine control systems.
- Developed a robust framework for the certification process, ensuring that all necessary documentation and testing procedures were in place.
- Implemented a rigorous testing and validation process to ensure that all control systems met the required standards and specifications.
- **Wind Farm Control:** Developed and implemented advanced control algorithms for wind farm optimization, resulting in improved energy production and reduced maintenance costs.
- Conducted extensive simulations and modeling to evaluate the performance of control strategies under various operating conditions.
- Collaborated with cross-functional teams to integrate control systems with existing infrastructure and ensure seamless operation.
- **Performance Validation:** Developed a method to quantify the power benefit of wake steering in offshore wind farms using only existing SCADA data, without the need for external measurement tools.
- Implemented uncertainty quantification and optimization techniques to improve the accuracy and reliability of the wake steering power benefit measurements.
- **Certification:** Developed a comprehensive understanding of the IEC 61400 standard and its application to wind turbine control systems.
- Implemented a rigorous testing and validation process to ensure that all control systems met the required standards and specifications.
- **Collaboration and Communication:** Collaborated with cross-functional teams, including Solution Owners, Product Owners, and Tool Developers, to ensure that all aspects of the certification process were aligned and integrated.
- Communicated complex technical concepts and findings to both technical and non-technical stakeholders, ensuring that all parties had a clear understanding of the certification process and its outcomes.

Industry Mention @ IEA Wind Task 44

An industry-oriented method to measure Wake Steering Power Benefit in Offshore wind farms | *k-Sahu/go-salesfo*

- Long-term validation using only existing SCADA data without external measurement tools.
- Uncertainty quanitfication and optimization of wake steering power benefit.

Education & Certifications

University of Central Florida *Jun 2019*
B.S. in Computer Science

Salesforce Certifications
Platform Developer II, Platform Developer I, Platform App Builder, Administrator